WEEK- 3

1) Factorial

```
haasini@LAPTOP-JG5OMVC7:

  GNU nano 8.7.1                                              factorial.c
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char *argv[]) {
    int num, i;
    long long fact = 1;

    if (argc != 2) {
        printf("Usage: ./factorial <number>\n");
        return 1;
    }

    num = atoi(argv[1]);

    for (i = 1; i <= num; i++)
        fact *= i;

    printf("\n--- Child Process (exec) ---\n");
    printf("Factorial of %d = %lld\n", num, fact);

    return 0;
}
```

Output:

```
haasini@LAPTOP-JG5OMVC7:~$ mkdir factorial
cd factorial
mkdir: factorial: File exists
haasini@LAPTOP-JG5OMVC7:~/factorial$ nano factorial.c
haasini@LAPTOP-JG5OMVC7:~/factorial$ gcc factorial.c -o factorial
haasini@LAPTOP-JG5OMVC7:~/factorial$ ./factorial 5

--- Child Process (exec) ---
Factorial of 5 = 120
haasini@LAPTOP-JG5OMVC7:~/factorial$ |
```

2) Parent Process

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    int num;
    pid_t pid;
    char str[20];

    printf("Enter a number: ");
    scanf("%d", &num);

    pid = fork();

    if (pid < 0) {
        printf("Fork Failed\n");
        exit(1);
    }

    if (pid == 0) {
        // Child Process
        sprintf(str, "%d", num);

        execl("./factorial", "factorial", str, NULL);

        // Executes only if exec fails
        perror("exec failed");
    }
    else {
        // Parent Process
        wait(NULL);

        printf("\n--- Parent Process ---\n");
        printf("Square of %d = %d\n", num, num * num);
    }

    return 0;
}
```

Output:

```
haasini@LAPTOP-JG5OMVC7:~$ mkdir factorial
cd factorial
mkdir: factorial: File exists
haasini@LAPTOP-JG5OMVC7:~/factorial$ nano factorial.c
haasini@LAPTOP-JG5OMVC7:~/factorial$ gcc factorial.c -o factorial
haasini@LAPTOP-JG5OMVC7:~/factorial$ ./factorial 5

--- Child Process (exec) ---
Factorial of 5 = 120
haasini@LAPTOP-JG5OMVC7:~/factorial$ nano factorial.c
haasini@LAPTOP-JG5OMVC7:~/factorial$ cd ~
mkdir parent_process
cd parent_process
haasini@LAPTOP-JG5OMVC7:~/parent_process$ nano parent_process.c
haasini@LAPTOP-JG5OMVC7:~/parent_process$ execl("./factorial", ..
-bash: syntax error near unexpected token `"./factorial",'
haasini@LAPTOP-JG5OMVC7:~/parent_process$ cp ~/factorial/factoria
haasini@LAPTOP-JG5OMVC7:~/parent_process$ gcc parent_process.c -o
haasini@LAPTOP-JG5OMVC7:~/parent_process$ ./parent_process
Enter a number: 5

--- Child Process (exec) ---
Factorial of 5 = 120

--- Parent Process ---
Square of 5 = 25
haasini@LAPTOP-JG5OMVC7:~/parent_process$
```